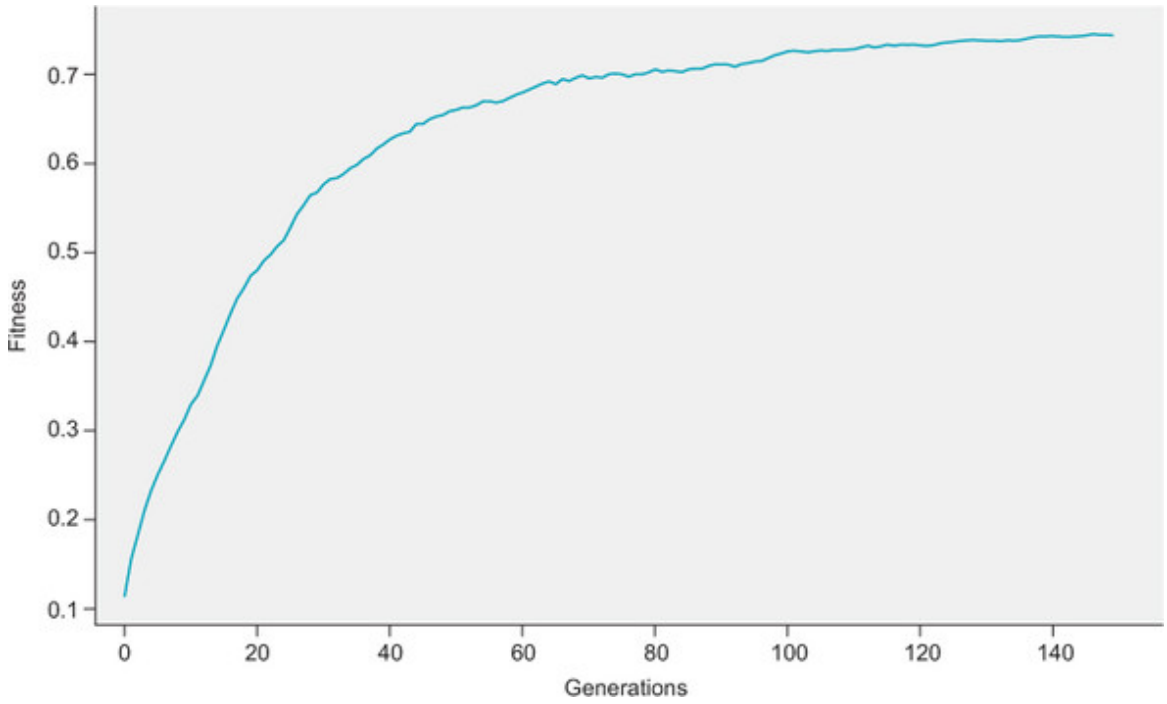


When we're optimizing real-valued parameters in a model, even a small increase in value might improve the fitness, and we can exploit that, which makes optimization faster. But although discrete-valued individuals are harder to optimize in an evolutionary algorithm, they are *impossible* to optimize using vanilla gradient descent and backpropagation, because they are not differentiable.

6.3. A genetic algorithm for CartPole

Let's see how this evolution strategy works in a simple reinforcement learning example. We're going to use an evolutionary process to optimize an agent to play CartPole, the environment we introduced in chapter 4 where the agent is rewarded for keeping the pole upright (figure 6.7).

Figure 6.7. We will use the CartPole environment to test our agent. The agent is rewarded by keeping the pole upright, and it can move the cart left or right.